Roll No. | | | | | | | |

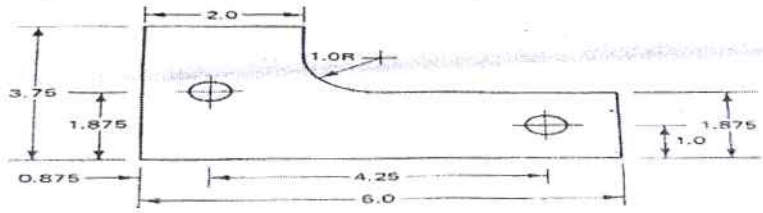
END SEMESTER EXAMINATION - 2024

Name of the Course: M.Tech (CAD/CAM Engineering) Semester: 1st
Name of the Paper: Computer Aided Manufacturing Paper Code: MME-103

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20marks)	CO1
(a)	Contrast open-loop and closed-loop systems in CAM.	
(b)	Evaluate the necessity and future prospects of NC systems and CAM, emphasizing their benefits and limitations.	
(c)	Distinguish between conventional machine tools and NC machine tools. Discuss various strategies for enhancing accuracy and productivity.	
Q2	(20 marks)	CO2
(a)	Compare different types of CNC machine tool system devices, such as encoders and interpolators.	
(b)	Explain computer-based process monitoring, control systems, and adaptive control systems in detail.	
(c)	Outline the key features of CNC systems. Discuss Direct Numerical Control (DNC), standard controllers, and the common programming functionalities offered by CNC systems.	
Q3	(20 marks)	CO3
(a)	Discuss computer-aided NC programming using the APT language. Provide an overview of manual programming for basic operations such as turning, milling, and drilling.	
(b)	Categorize the different components of an NC program. Explain the process of generating NC programs through CAD/CAM systems using an example geometry.	
(c)	Develop an APT geometry NC part program to define the part outline for a milling operation based on the geometry shown in Fig. 1.  (Fig. 1)	
Q4	(20 marks)	CO4
(a)	Enumerate Computer Aided Process Planning Methods. Compare Manual and Computer Aided Process Planning.	
(b)	Examine variant and generative process planning methods, providing a detailed explanation of each.	
(c)	List the methods used in Computer-Aided Process Planning (CAPP) and compare manual process planning with computer-aided approaches.	

Q5	(20 marks)	CO5/CO6
(a)	Describe the key features, components, and applications of Computer Integrated Manufacturing.	
(b)	Discuss the various elements of expert systems and artificial intelligence in manufacturing, highlighting the applications of neural networks in this domain.	
(c)	List the advantages and disadvantages of Computer Integrated Manufacturing (CIM).	